+91 7680830711
gask402@gmail.com
kamal-manchenella
Darkxzie
Kamel

Kamal Manchenella

Education

2024 – Present — **B.E. in Mathematics and Computing**, *BITS Pilani, Goa Campus*, India
Currently in 2nd Year

2022 – 2024 — **Senior Secondary Education (Class XII)**, *Delhi Public School, Hyderabad*, India
Graduated with a focus on Science and Mathematics

Work Experience

May 2026 – Present — **AI Workflows Intern**, *Swecha*
Developing AI-assisted workflows and automation tools for open-source language-data initiatives. Building Python-based corpus-processing and token-level ingestion pipelines while integrating command-line tools for reliable and scalable data collection.

Apr 2026 – Present — **Research Intern**, *Deep Forest Sciences*
Working on DeepRetro, a hybrid retrosynthesis planning framework integrating machine learning and rule-based methods for automated reaction pathway generation.

Dec 2025 – Present — **Student Researcher**, *APPCAIR (Applied AI Research Center), BITS Pilani*
Working under Prof. **Tanmay Tulsidas Verlekar** on peer-review rebuttal alignment in collaboration with **TCS Research**.

Aug 2025 – Nov 2025 — **Student Researcher**, *Pragya*
Contributed to ongoing research in Large Language Models under Prof. **Amitava Das** through the internship program.

Volunteering

Aug 2025 – Jan 2026 — **Campus Partner**, *Perplexity*
Supported Perplexity's campus initiatives by promoting its AI products, coordinating student outreach, and increasing awareness across the campus community.

Aug 2025 – Jan 2026 — **Instructor**, *Center for Technical Education, BITS Goa*
Taught an intermediate course on *Building LLMs from Scratch*, covering transformers, training pipelines, fine-tuning, and optimization of modern large language models.

Projects

Robust Loss Functions — Explored and implemented CE, FL, NCE, NFL, and APL loss functions under varying noise rates. Built CNNs from scratch and evaluated model performance across 40% noise-injected CIFAR-10 data.

Diffusion — Implemented a diffusion-based generative model trained on CIFAR-10. Investigated the impact of classifier-free guidance, sampling steps, and attention mechanisms on generation quality using FID and CMMD scores.

Obesity Risk Prediction Implemented and compared ensemble methods including Random Forest, Gradient Boosting, and XGBoost on structured health data. Achieved approximately 96% classification accuracy using a voting classifier.

Neural Net from Scratch Built a fully connected neural network using only NumPy to classify handwritten digits from the MNIST dataset. Trained the network using stochastic gradient descent and achieved 97.8% accuracy.

Technical Reports

Apr 2025 **ML and Diffusion Report** – Authored a detailed analysis of custom loss functions for noisy CIFAR-10 classification and diffusion-model performance. Covered model architectures, evaluation under label noise, and metrics including FID and CMMD.

Jun 2025 **Ensemble Learning and Neural Net Report** – Wrote an in-depth report covering ensemble learning for obesity-risk prediction and a neural-network implementation for MNIST. Emphasized performance comparisons and hands-on architecture design.

Clubs & Societies

Jun 2025 – Present **Member**, *Society for Artificial Intelligence and Deep Learning (SAiDL)*, Goa, India
Involved in machine-learning research, technical assignments, reading groups, and peer collaboration.

Aug 2024 – Aug 2025 **Marketing Associate**, *Department of Sponsorship and Marketing, BITS Goa*, Goa, India
Responsible for cold calling, sponsor outreach, and promotional strategy.

Aug 2024 – Sep 2025 **Junior Developer**, *Developers' Society, BITS Goa*, Goa, India
Worked with C++, Python, and machine-learning tools on society projects and technical assignments.

Skills

Languages C, C++, Python, Java

Machine Learning PyTorch, NumPy, scikit-learn, XGBoost

Tools Git, GitHub, GitLab, LaTeX

Other Research, Technical Writing, Event Planning, Marketing, Remote Collaboration

Certifications

Mar – Sep 2025 **Machine Learning Specialization**, DeepLearning.AI and Stanford University – Covered supervised learning, unsupervised learning, reinforcement learning, model evaluation, and practical machine-learning system design.

Aug – Sep 2025 **Deep Learning Specialization**, DeepLearning.AI – Covered neural networks, CNNs, RNNs, sequence models, hyperparameter tuning, and model optimization.